

A: Datasheet

Algorithm: afrengine_001

Developer: AFR Engine

Submission Date: 2024_03_15

Template size: 2048 bytes

Template time (2.5 percentile): 820 msec

Template time (median): 974 msec

Template time (97.5 percentile): 994 msec

Investigation:

Frontal mugshot ranking 140 (out of 481) -- FNIR(1600000, 0, 1) = 0.0017 vs. lowest 0.0008 from intema_001

Mugshot webcam ranking 168 (out of 443) -- FNIR(1600000, 0, 1) = 0.0125 vs. lowest 0.0054 from sensetime_009

Mugshot profile ranking 74 (out of 412) -- FNIR(1600000, 0, 1) = 0.0709 vs. lowest 0.0510 from nec_009

Immigration visa-border ranking 238 (out of 370) -- FNIR(1600000, 0, 1) = 0.0099 vs. lowest 0.0005 from psl_001

Immigration visa-kiosk ranking 126 (out of 314) -- FNIR(1600000, 0, 1) = 0.0730 vs. lowest 0.0387 from cloudwalk_mt_002

Identification:

Frontal mugshot ranking 66 (out of 481) -- FNIR(1600000, T, L+1) = 0.0028, FPIR=0.003000 vs. lowest 0.0010 from sensetime_009

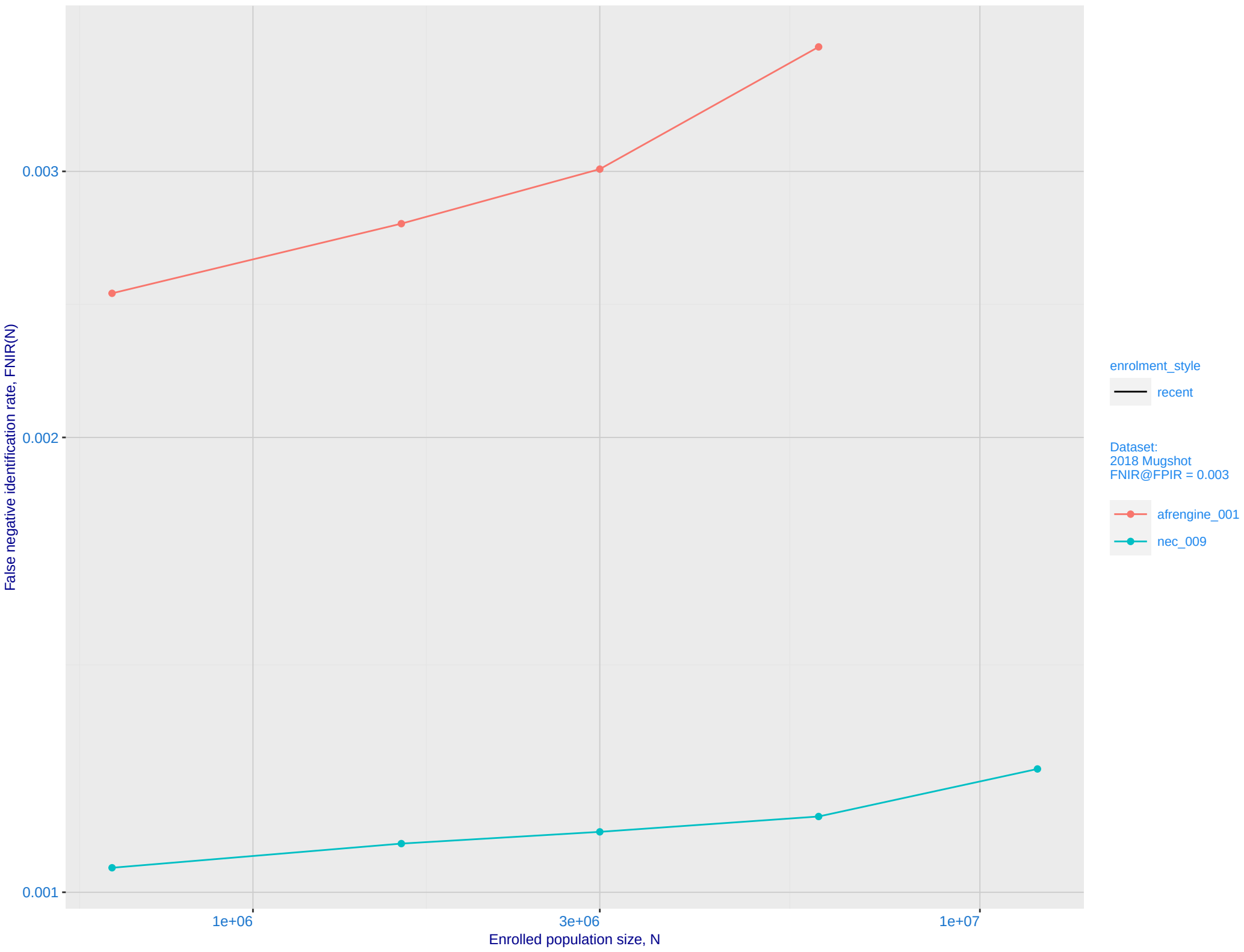
Mugshot webcam ranking 107 (out of 441) -- FNIR(1600000, T, L+1) = 0.0254, FPIR=0.003000 vs. lowest 0.0065 from sensetime_009

Mugshot profile ranking 35 (out of 411) -- FNIR(1600000, T, L+1) = 0.1577, FPIR=0.003000 vs. lowest 0.0591 from cloudwalk_mt_002

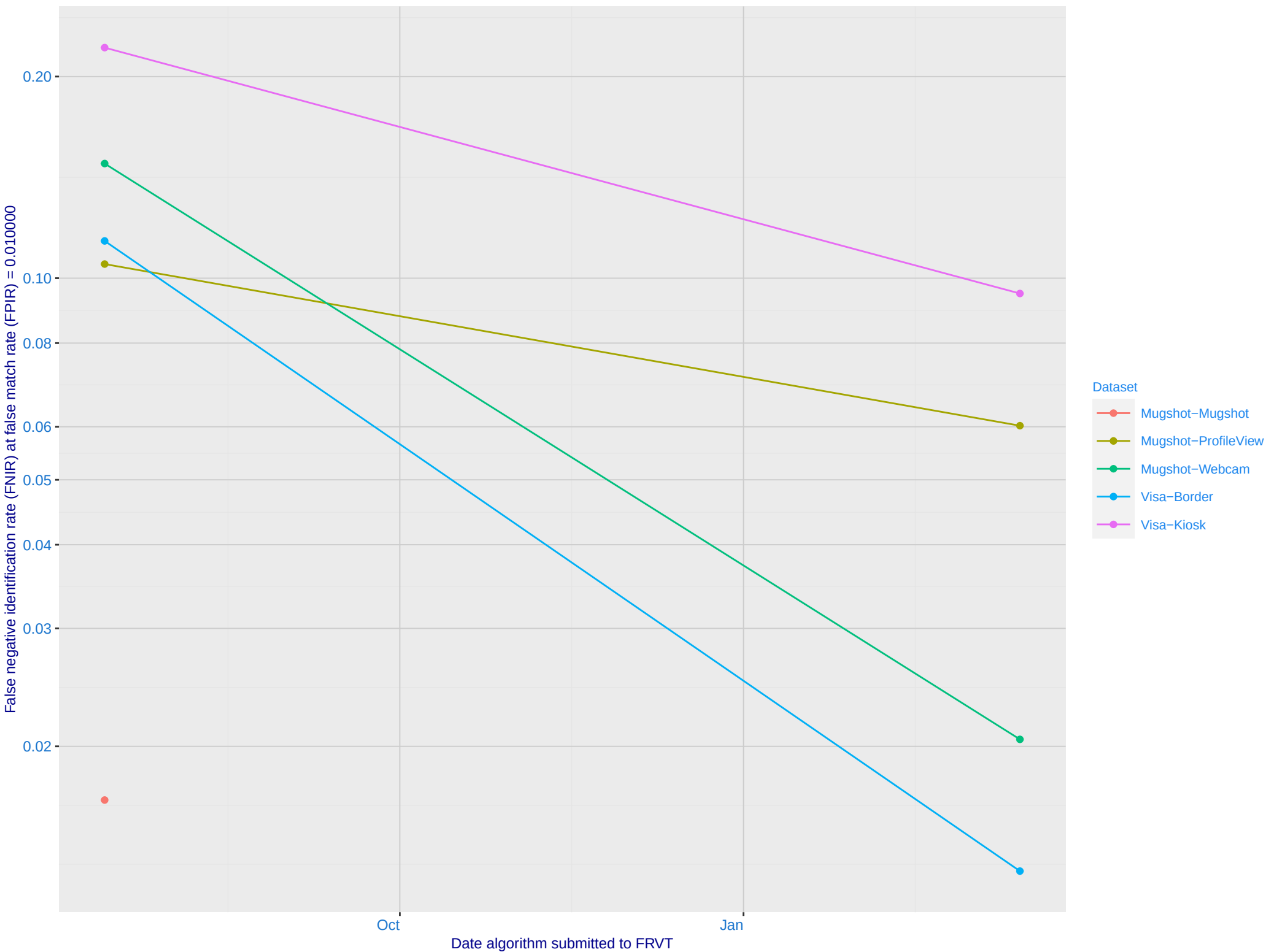
Immigration visa-border ranking 143 (out of 369) -- FNIR(1600000, T, L+1) = 0.0150, FPIR=0.003000 vs. lowest 0.0009 from cloudwalk_mt_002

Immigration visa-kiosk ranking 86 (out of 314) -- FNIR(1600000, T, L+1) = 0.1048, FPIR=0.003000 vs. lowest 0.0460 from nec_009

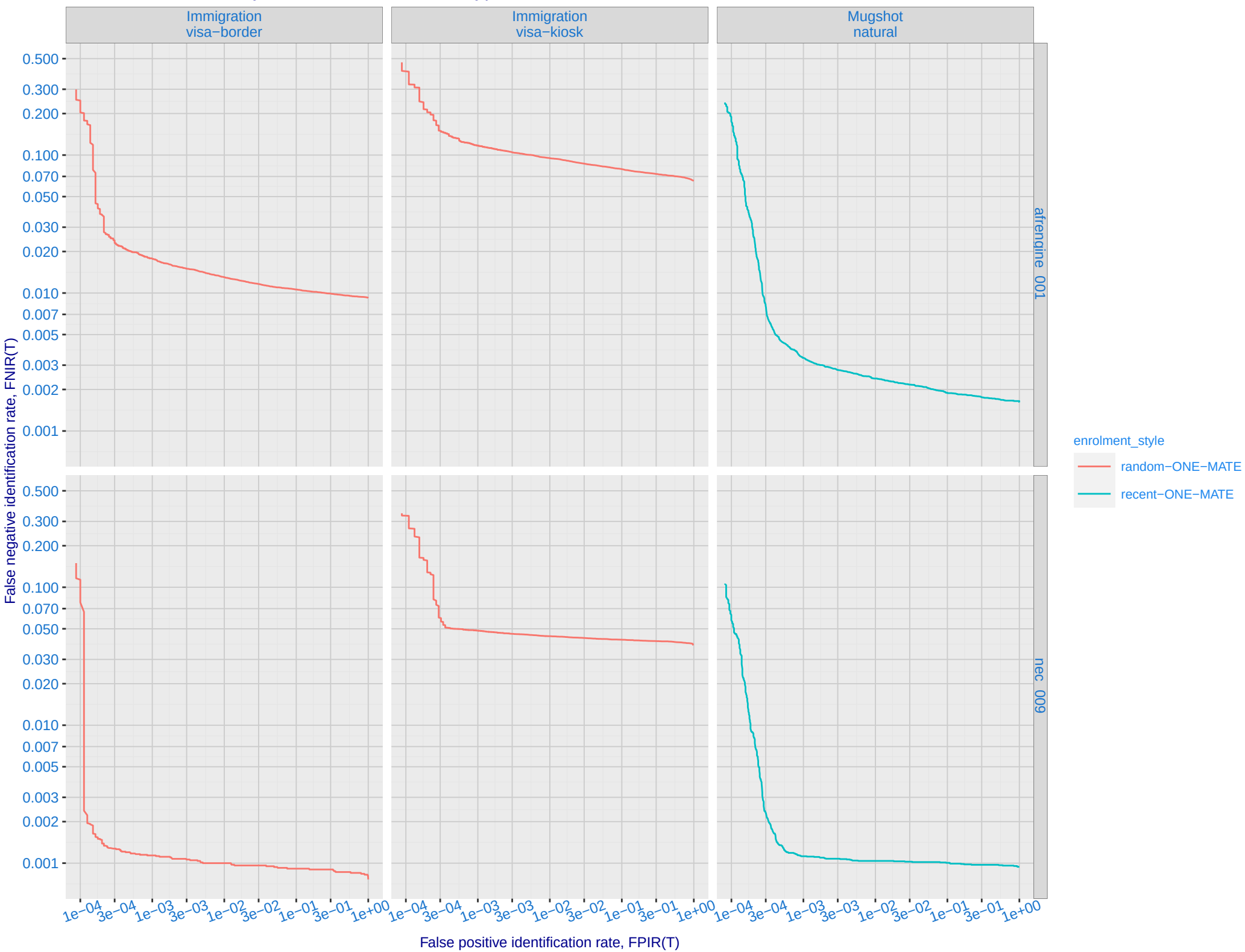
B: Mugshot natural images, identification mode: FNIR(N, L+1, T) vs. most accurate (nec_009)



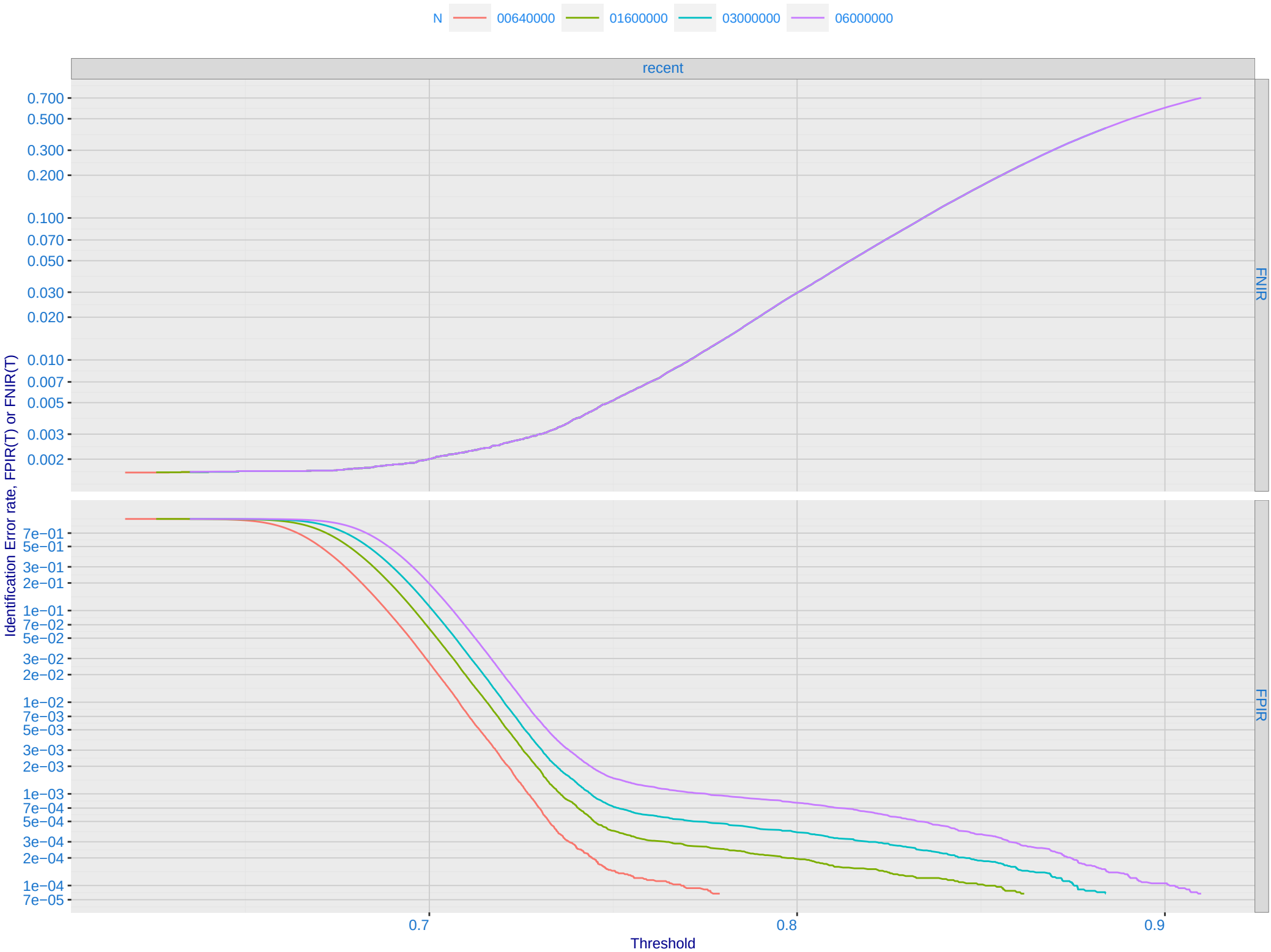
C: Evolution of accuracy for AFRENGINE algorithms on three datasets 2018 – present



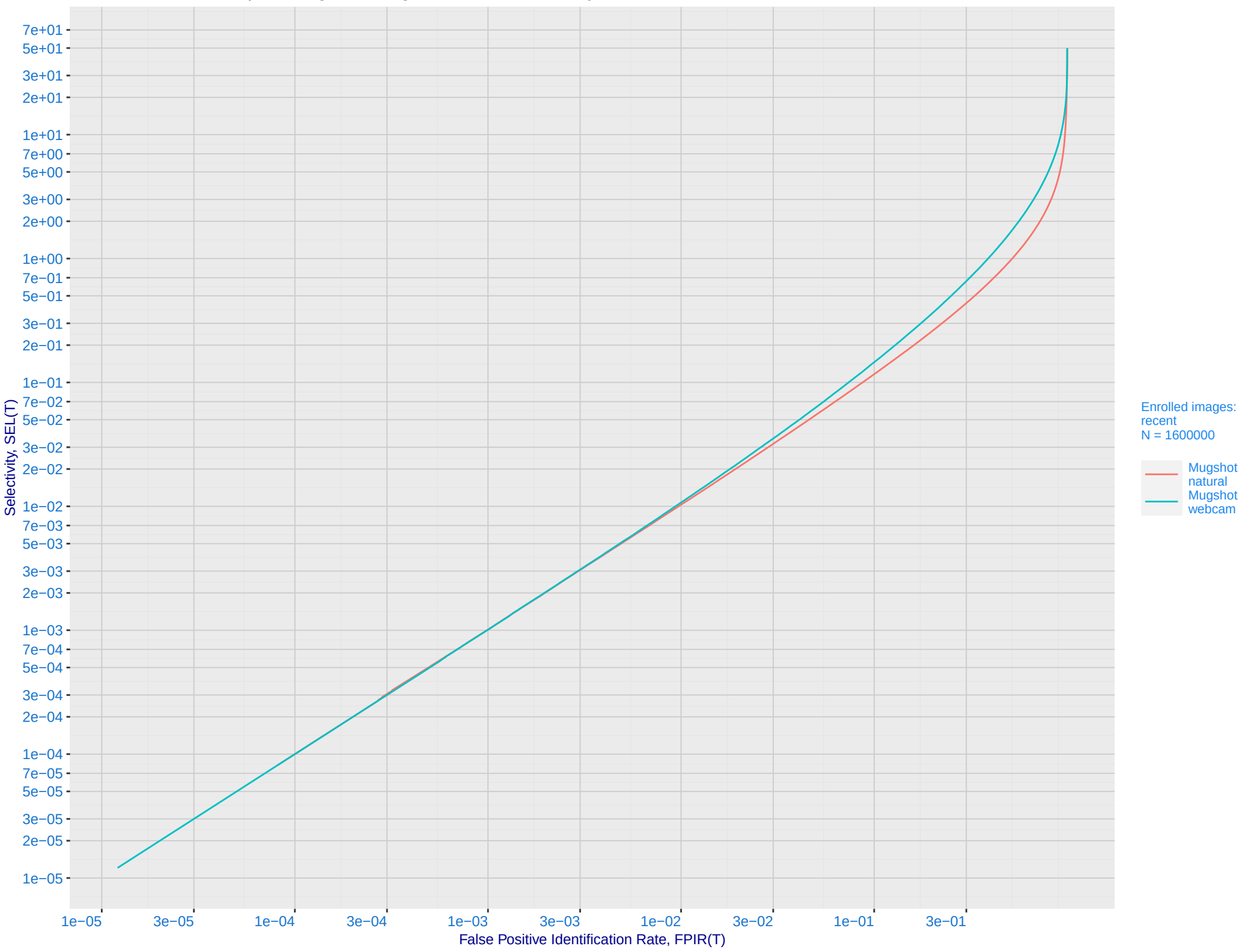
D: 1:N error tradeoff by dataset and enrollment type. N = 1600000 individuals



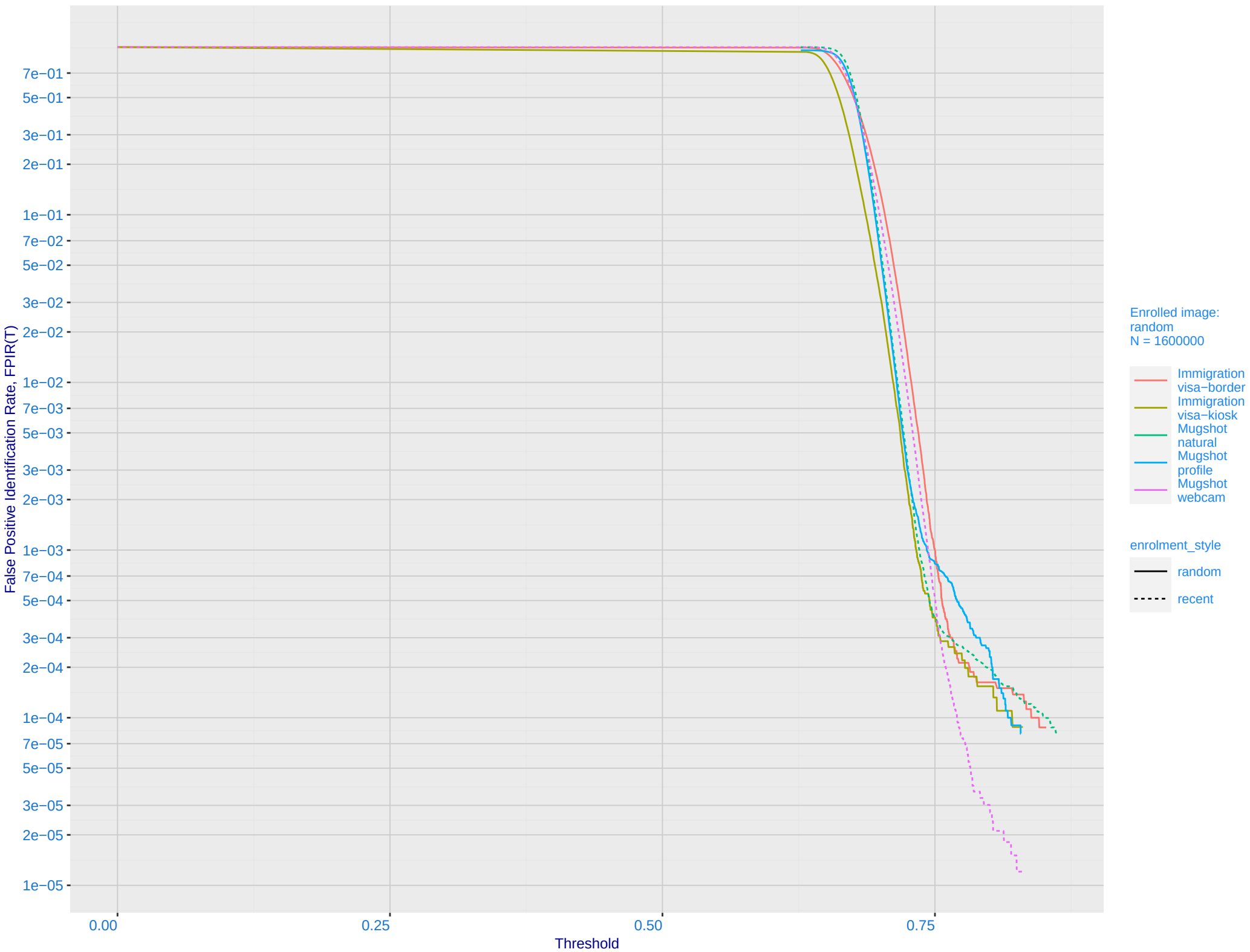
E: Dependence of error rates on T by number enrolled identities, N, for Mugshot natural images



F: FPIR vs. Selectivity for mugshot images, N = 1600000 subjects enrolled with one recent mate

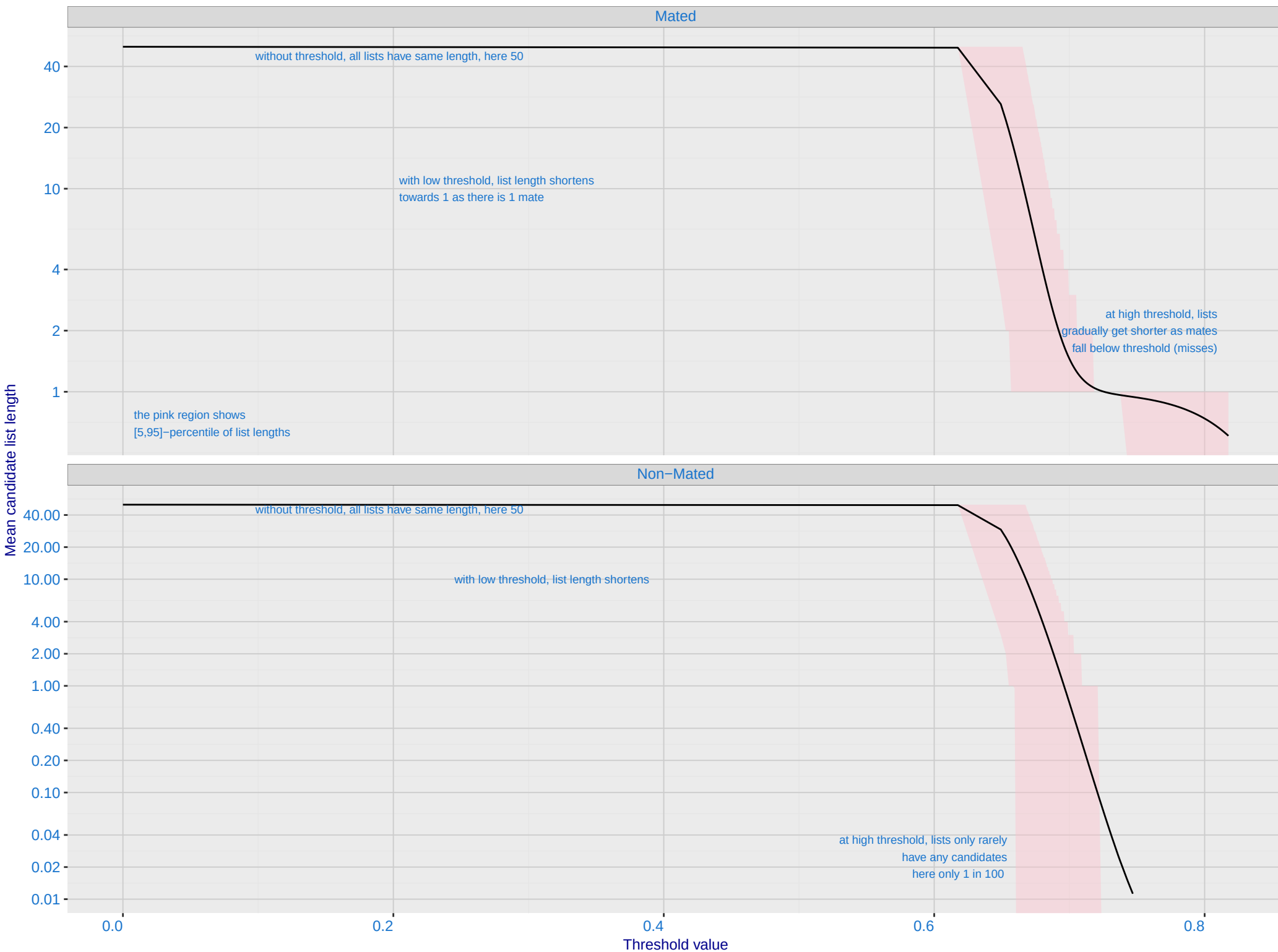


G: FPIR dependence on T by probe type for N = 1600000 subjects



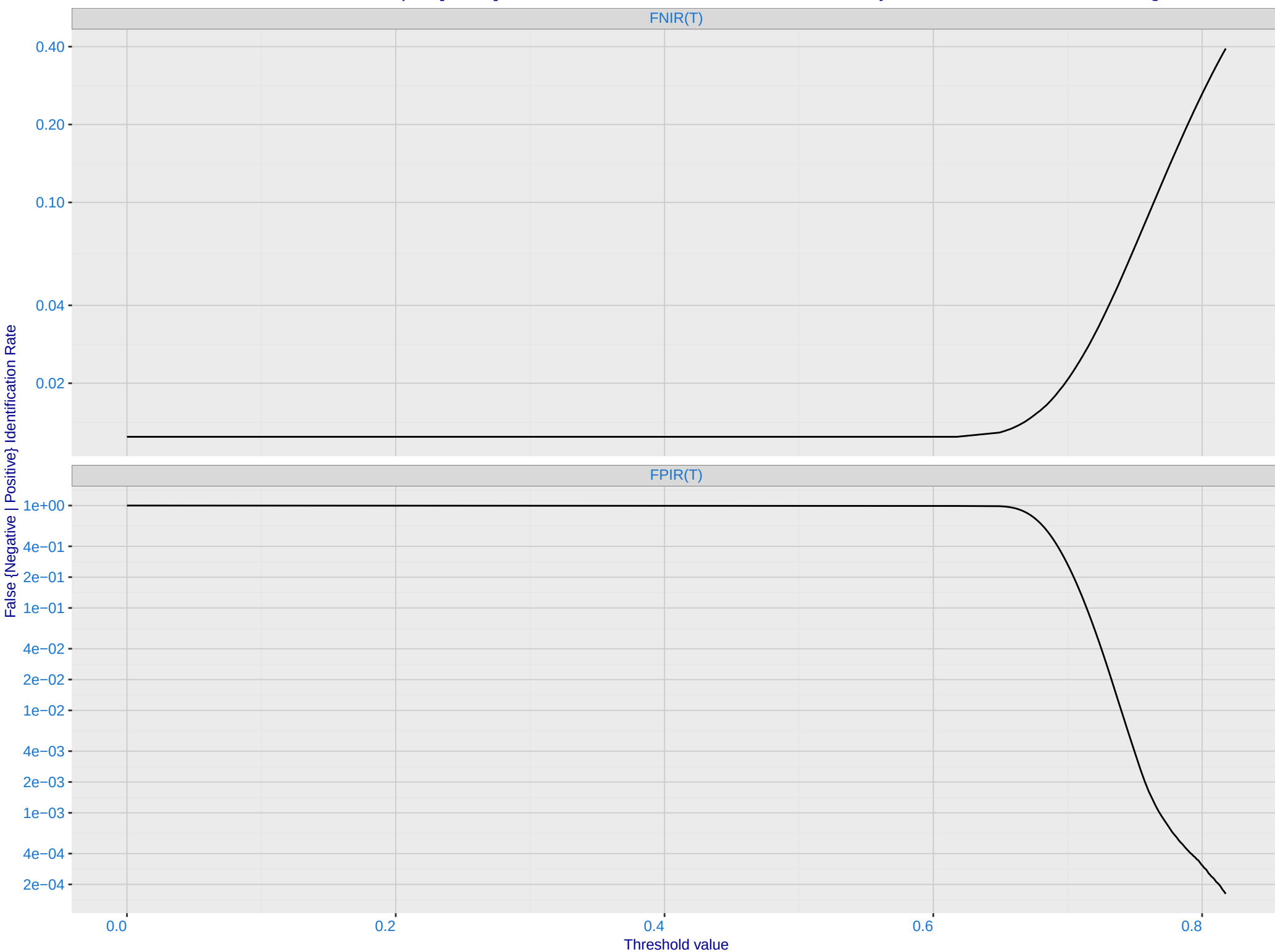
H: Reduced length candidate lists for human review

Dataset is border-border with time-lapse [10,15] YRS with N = 1600000. Probes are 10-15 years later than enrollment image

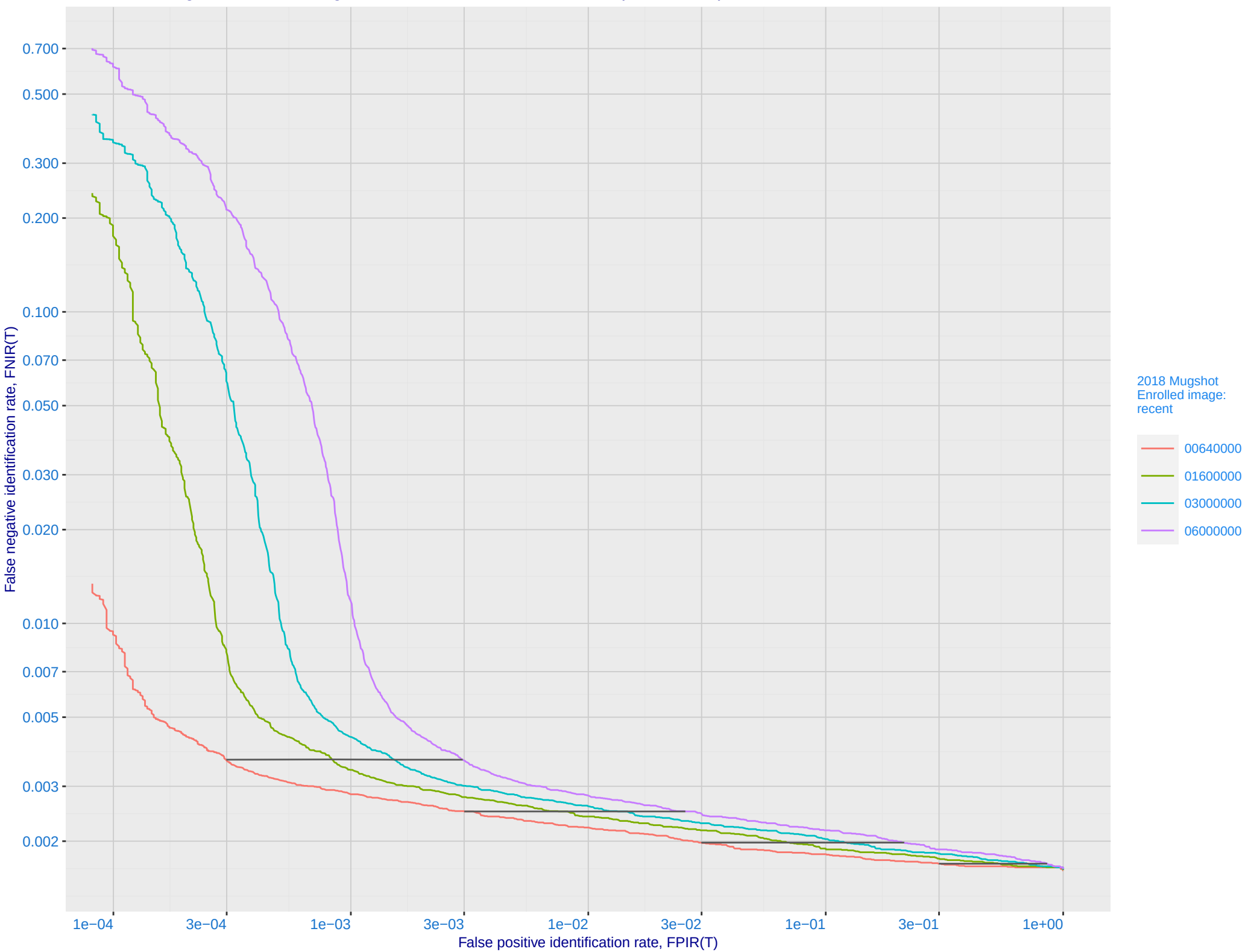


I: FNIR and FPIR dependence on threshold

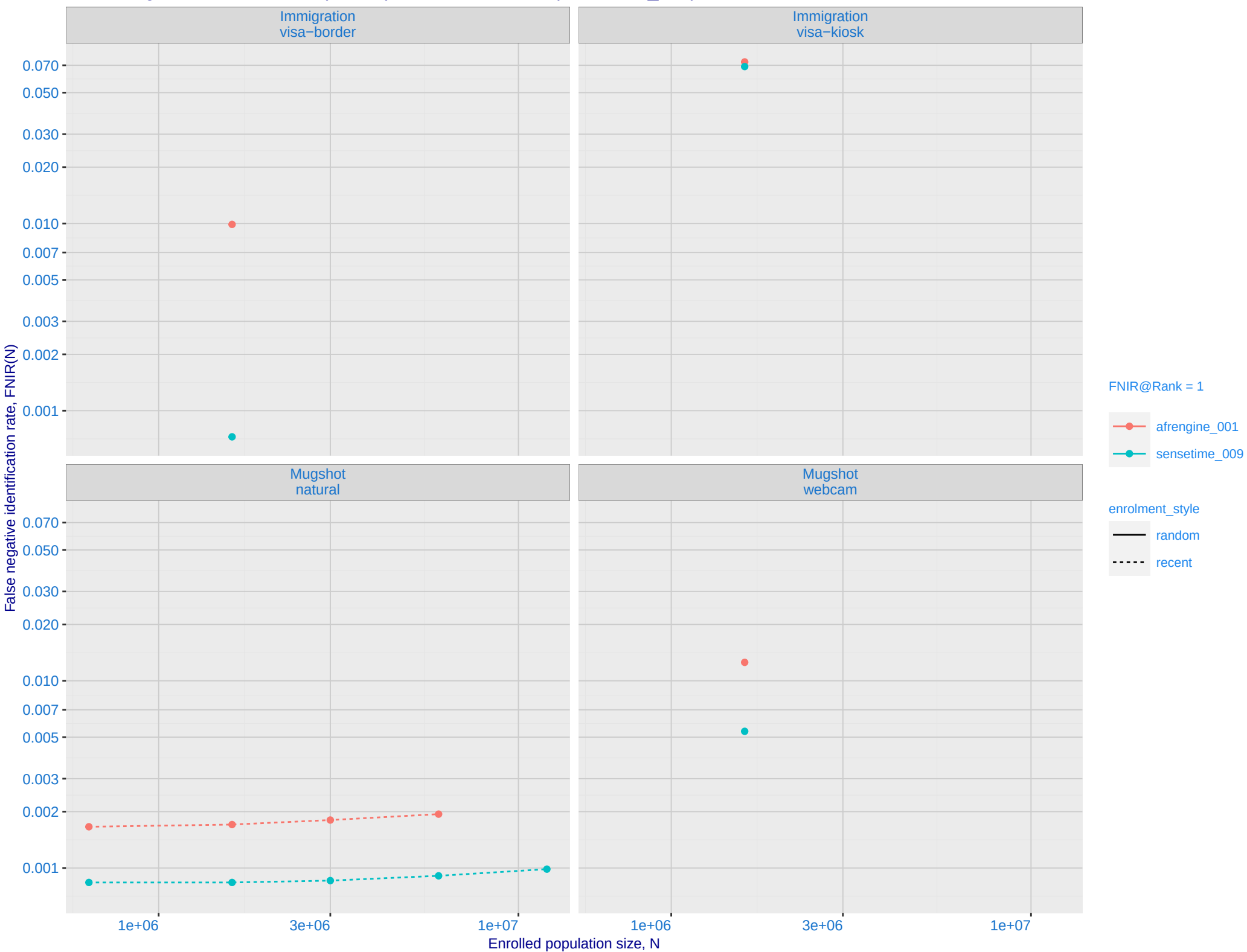
Dataset is border-border with time-lapse [10,15] YRS with N = 1600000. Probes are 10-15 years later than enrollment image



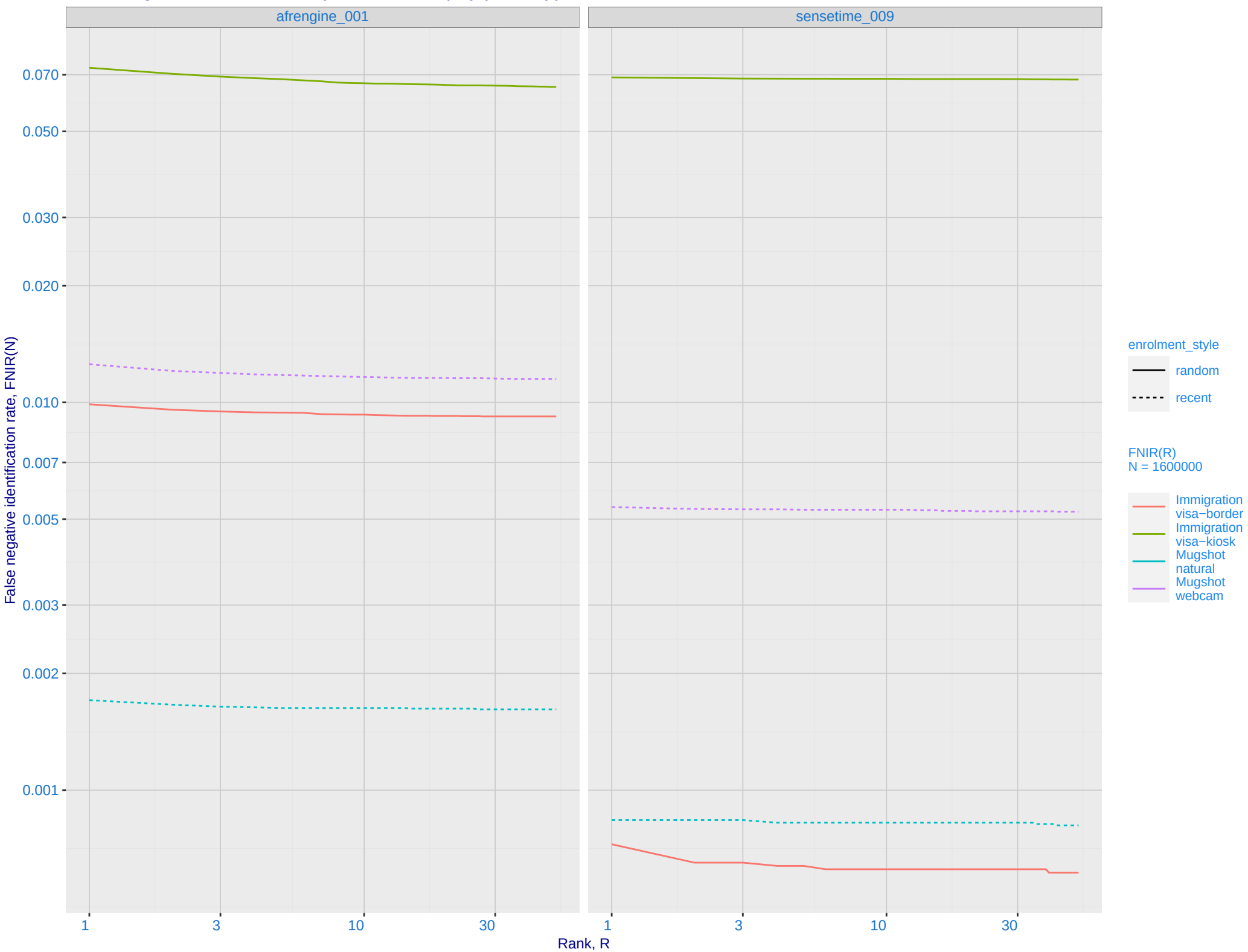
J: DET for Mugshot natural images and various N. Links connect points of equal threshold.



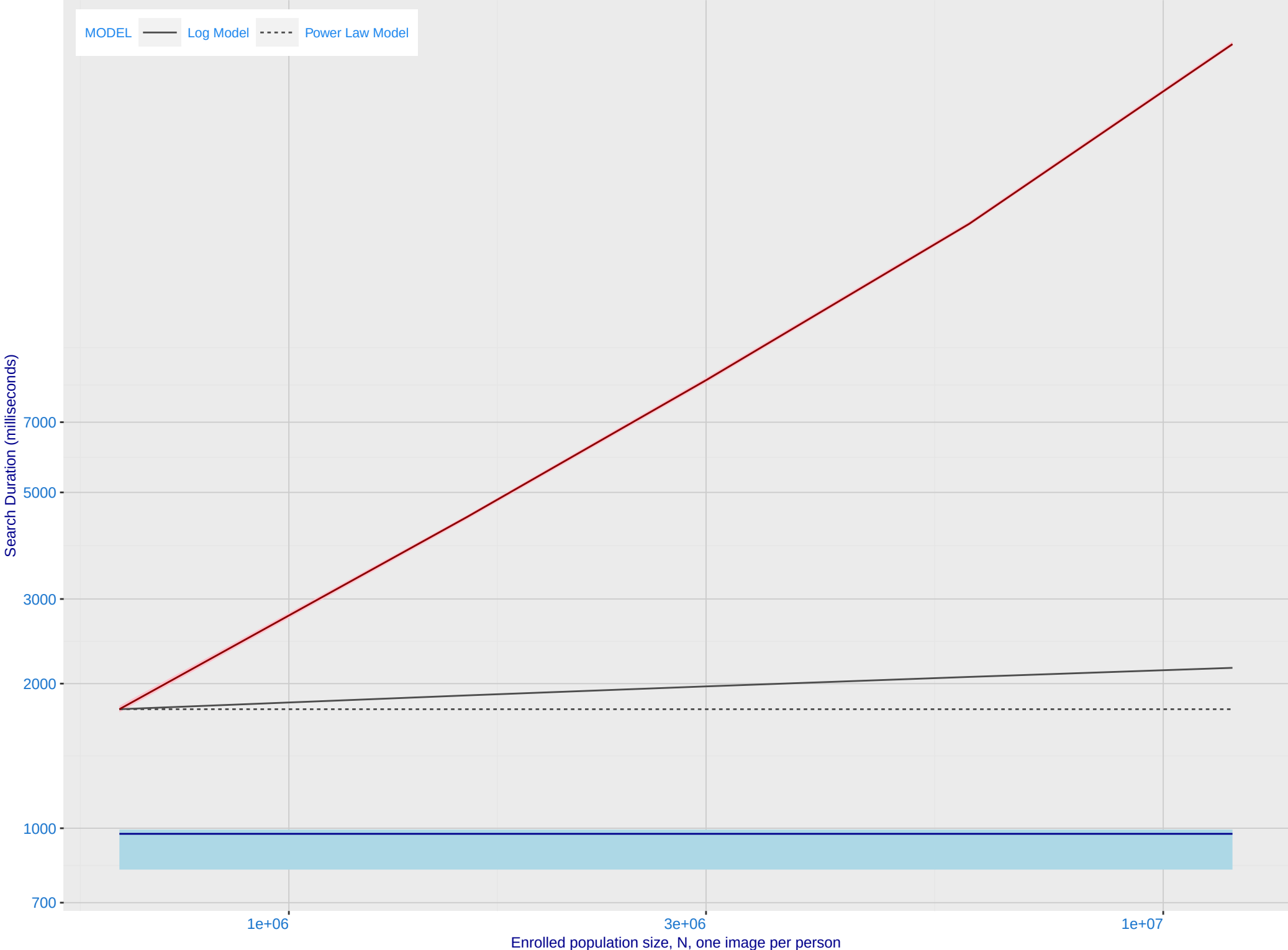
K: Investigational mode: FNIR(N, 1, 0) vs. most accurate (sensetime_009)



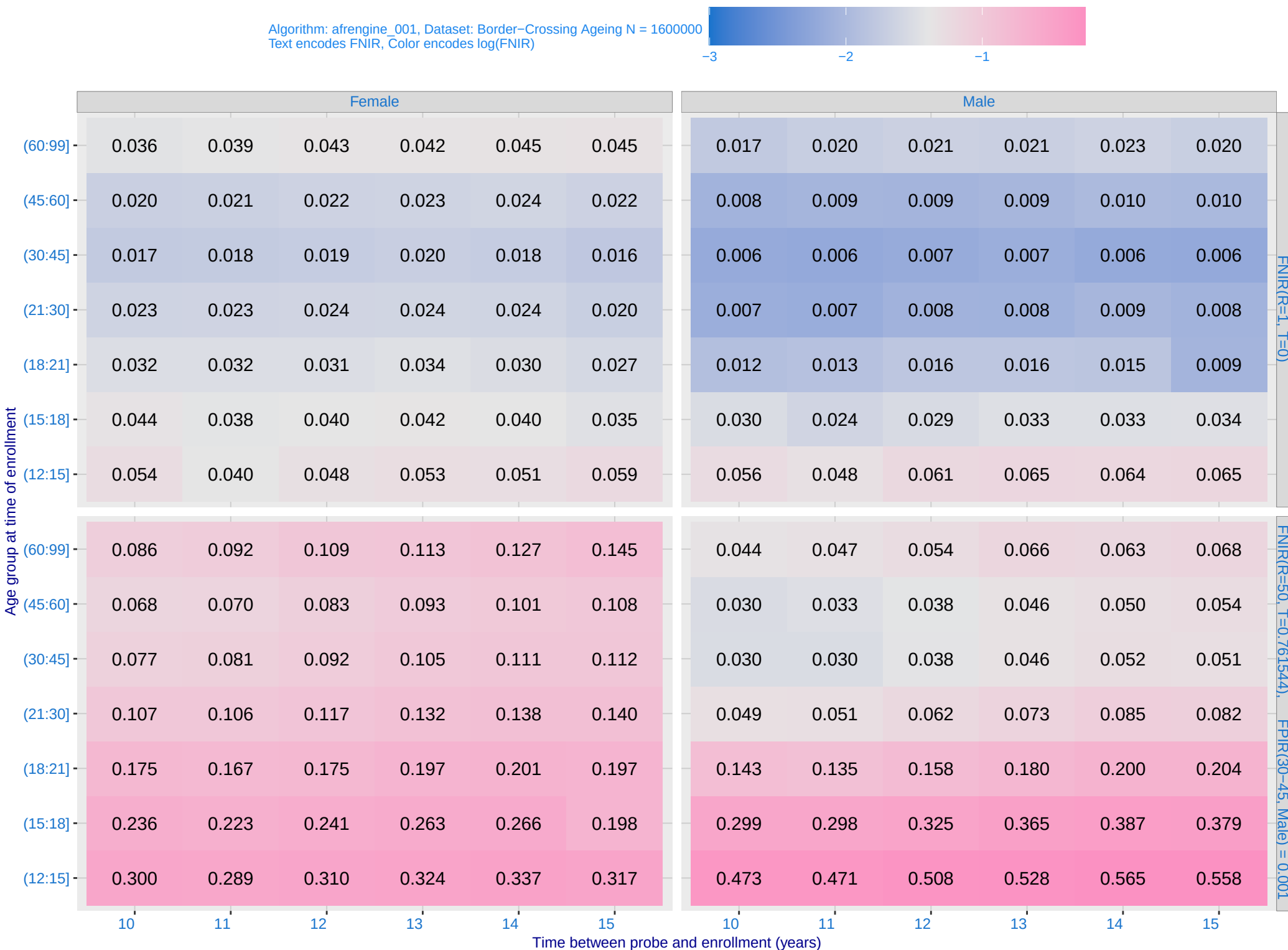
L: Investigational mode: FNIR(1600000, R, 0) by probe type



M: Template duration; search duration vs. N. The blue and pink ribbon covers 95 percent of observed measurements. The template generation time is independent of N. The log and power-law models are fit to the first two (N,T) observations



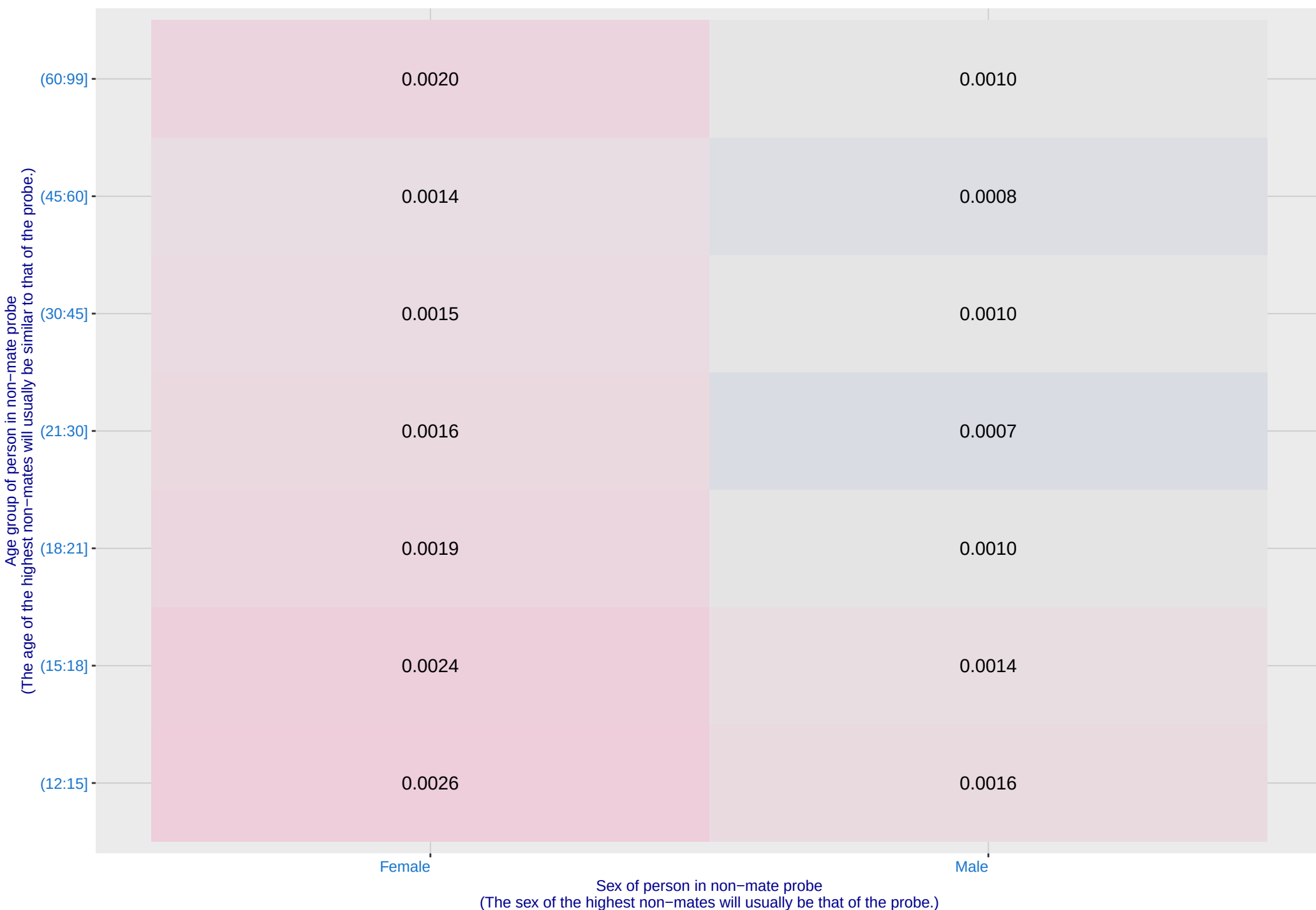
O: FNIR(T, N = 1.6 million) by sex, age and time-lapse. The top row gives investigational rank-1 miss rates. The bottom panels give high threshold for more lights-out identification with low FPIR.



P: FPIR(N = 1.6 million) by sex and age. It is typical for false positive identification rates to be higher in women except in their teens.

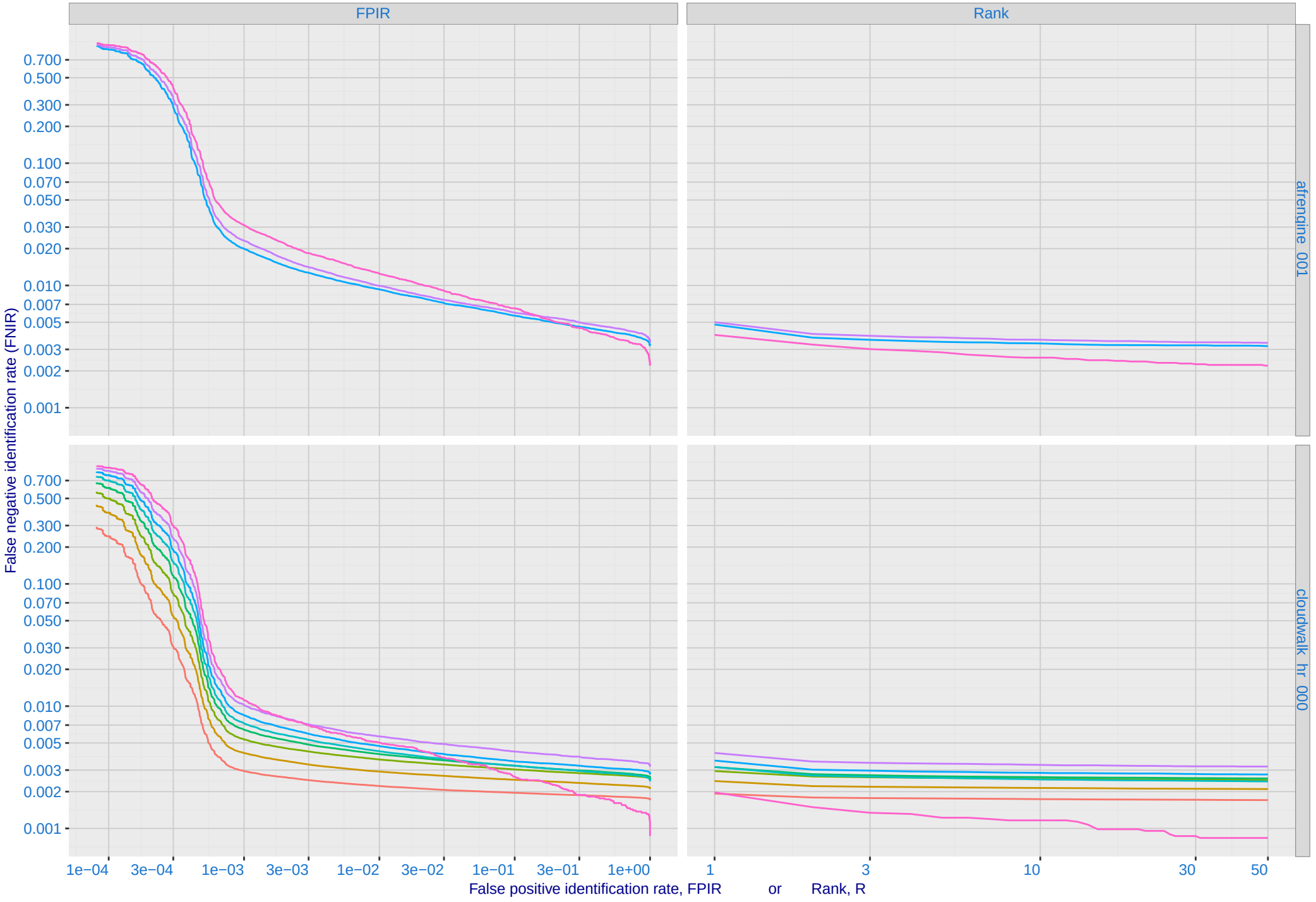
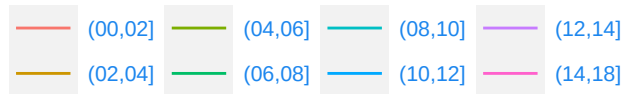
Algorithm: afrengine_001, Dataset: Border-Crossing Ageing
Threshold: 0.761544 set to achieve FPIR(30-45, Male) = 0.001

Color encodes log(FPIR)



Q: Identification FNIR(N, T, L+1) and Investigational FNIR(N, 0, R) under ageing

Dataset: 2018 Mugshot N = 3068801



R: Decline of genuine scores with ageing, with some eventually dropping below typical thresholds shown by the horizontal lines

